

PALLETPILOT · PROBLEM STATEMENT

PalletPilot problem statement

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DOCUMENT CONTROL

Document ID	PLP-PRB-001	Version	0.5
Status	Draft	Date	2026-10-02
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/palletpilot		
Source	docs/01-problem.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.5	2026-10-02	Amish Chadha	Follow mode classification, donor survey and first candidate partners decided by Amish on 2026-10-02 (PLP-DEC-001)
0.4	2026-09-26	Amish Chadha	Budget approved by Amish (\$1,610, PLP-DDR-002)
0.3	2026-09-25	Amish Chadha	Record decisions PLP-DDR-001 (budget \$1,550, layered stopping, research-use legal route, partner types); OSHA wording checked against the source
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (problem, users, context, constraints, out of scope, prior work with sources)
0.1	2026-09-24	Amish Chadha	Initial scaffold

PalletPilot problem statement

Small warehouses, shops and workshops move pallets with manual pallet jacks, and a loaded jack takes more force to start and keep rolling than ergonomic limits allow for repeated shifts. Powered walkie pallet jacks now cost from about \$1,640, but trucks that follow the operator hands-free are forklift-class products well beyond a small business. PalletPilot is an open retrofit kit that adds powered drive, a follow-me mode and layered stopping to the manual jack the site already owns.

■ The problem

A manual pallet jack is the most common load mover in small facilities. A typical unit has 685 x 1220 mm (27 x 48 in) forks, a 2,500 kg (5,500 lb) rating, a 75 mm (3 in) lowered height and a mass of about 64 kg (140 lb), and costs about \$396 new ([Global Industrial](#)).

Moving it loaded is hard work:

- **Force.** With a 1,000 kg pallet, the rolling force is about 130 N (13 kgf) and the force to start moving is about 270 N (28 kgf), assuming 1.2 % rolling resistance and 2.5 % breakaway resistance for polyurethane wheels on smooth concrete (estimates; to be measured). The widely used Liberty Mutual tables give maximum acceptable initial push forces of about 9 to 22 kg for 90 % of female workers, and sustained forces of about 4 to 14 kg, depending on handle height, distance and frequency ([Ciriello and Snook 1991, summarized by MHI](#)). A heavy pallet on a manual jack is at or past those limits every time it starts.
- **Walking time.** Order picking and staging are mostly walking. The operator pulls the jack, stops, walks back to pick, and pulls again. A truck that follows the operator removes the repeated grab, pull and release.

Three kinds of product exist, and none fits a small site well:

1. **Powered walkie pallet jacks.** A basic lithium walkie with 1,500 kg (3,300 lb) capacity now sells for about \$1,640 ([Home Depot](#), [Tory Carrier A-1034](#)). This solves the force problem but not the walking problem, and the site's existing manual jacks sit idle.
2. **Bolt-on power kits.** The PowerHandling PowerPallet 2000 converts a manual jack to powered drive in 10 to 20 minutes, with a 1,590 kg (3,500 lb) rating and a top speed of 1.65 m/s (3.7 mph), for about \$2,084 to \$2,294 ([HoF Equipment](#); [F.E. Bennett](#)). It proves the retrofit idea works but costs more than a new walkie and has no follow mode.
3. **Follow-me trucks.** Jungheinrich's easyPILOT Follow, shown at LogiMAT 2018, lets a low-level order picker follow an operator who carries a UWB control unit, with laser scanners for obstacle detection ([Jungheinrich](#)). It is an option on a full order-picking truck, not something a small business can add to a manual jack.

So a retrofit kit only earns its place if it does something a cheap walkie does not: follow-me operation, reuse of existing jacks and open, repairable parts. The kit's cost has to stay well below a new walkie plus a follow-me system.

■ Users and context

User	Need	Context
Small warehouse or 3PL operator	Move 20 to 100 pallets a shift without a forklift or a second worker	Single shift, one to five staff, concrete floor, some ramps
Retail backroom and grocery receiving	Stage deliveries and restock aisles, often one person working alone	Tight aisles, customers nearby, early-morning deliveries
Workshop, maker space or small manufacturer	Move raw stock and finished goods between machines	Mixed traffic, cables and debris on the floor
Order picker	Pick cartons onto a pallet while walking an aisle, without pulling the jack at every stop	Stop-start, short moves of 2 to 10 m
Maintenance person	Fit, inspect and repair the kit with ordinary tools and parts	No dealer service contract

Operating environment

- **Floors:** sealed or bare concrete, level, with expansion joints, dock plates and short ramps up to about 2 % grade. Dust, stretch-wrap scraps and small debris are common.
- **Loads:** pallets of 200 to 1,500 kg on 1,200 x 1,000 mm (48 x 40 in) stringer or block pallets.
- **Traffic:** people on foot, other jacks, occasionally a forklift; customers in retail backrooms.
- **Climate:** indoor, 0 to 40 °C; unheated storage and loading bays in winter.
- **Power:** ordinary 120 V or 230 V outlets for overnight charging.

■ Constraints

- Garage-buildable prototype, \$1,610 USD for the kit (budget raised from \$1,200 to \$1,550 by Amish on 2026-09-25, PLP-DDR-001, and to \$1,610 on 2026-09-26 to cover the priced BOM, PLP-DDR-002). The donor manual jack is not included.
- Bolt-on to a common 27 x 48 in manual jack with no welding, cutting or drilling of the jack's load-bearing structure, and removable to restore the jack.
- Keep the jack's 75 mm lowered height, fork length and its ability to enter a standard pallet.
- Walking pace only, with layered stopping as the pitch states: hardwired emergency stops, a non-contact lidar stop layer in follow mode, and a stop on bumper contact as the last layer.
- Low-voltage (24 V class) DC system with a lithium iron phosphate pack.
- Legal and workplace rules: in the United States a motorized pallet jack is a powered industrial truck, so every operator needs formal training and evaluation under 29 CFR 1910.178(l) ([OSHA 1910.178; summary](#)). The same standard states that "modifications and additions which affect capacity and safe operation shall not be performed by the customer or user without manufacturers prior written approval" (1910.178(a)(4), wording checked against the OSHA page on 2026-09-25). A retrofit on a jack in a workplace therefore needs the jack maker's approval. Amish decided on 2026-09-25 to document the kit for research use now and to approach a jack maker before any workplace trial (PLP-DDR-001, D8).
- A truck that moves with no one at the tiller comes close to the scope of ISO 3691-4 for driverless industrial trucks, which calls for personnel detection that stops the truck before it contacts a person ([ISO 3691-4](#);

[overview](#)). How a follow-me mode is classified must be settled before any trial with people. Decided 2026-10-02 (PLP-DEC-001): follow mode is treated as a driverless truck function under ISO 3691-4, with personnel detection and stopping functions to that standard, unless a standards body or a notified body confirms that the walking follower counts as the operator.

■ Out of scope

- Powered lifting. The donor jack's hand pump stays.
- Autonomous navigation, mapping or fleet management.
- Ride-on platforms or speeds above walking pace.
- Outdoor use, rough ground and steep ramps.
- Use in hazardous (explosive) atmospheres, cold stores below 0 °C and food-grade washdown areas.
- A commercial product. The first builds are research and demonstration prototypes.

■ Prior work

- **Manual pallet jacks.** Commodity 2,500 kg jacks at about \$396 set the donor platform and its dimensions ([Global Industrial](#)).
- **Low-cost lithium walkies.** About \$1,640 for a 1,500 kg, 24 V, 20 Ah walkie ([Home Depot](#)). This is the price ceiling that a retrofit kit has to beat.
- **Retrofit power kits.** The PowerPallet 2000 shows a clamp-on drive unit with a quick-change lithium pack and a truck mode at 0.67 m/s (1.5 mph) ([HoF Equipment](#)).
- **Follow-me order pickers.** easyPILOT Follow pairs UWB operator tracking with laser scanners ([Jungheinrich](#)). The laser scanners matter: UWB tells the truck where the operator is, not where anyone else is.
- **UWB ranging modules.** Qorvo's DWM3000 module ranges to about 10 cm precision on UWB channels 5 and 9 ([Qorvo](#)), which makes a low-cost operator tag practical.
- **Ergonomic push limits.** Ciriello and Snook's tables set the target for how much the kit must take off the operator ([MHI summary](#)).

■ Open questions

- Legal route for a workplace retrofit under 1910.178(a)(4): decided 2026-09-25 (PLP-DDR-001, D8): research use now, a jack maker's written approval before any workplace trial. Which jack maker to approach is open.
- Classification of follow-me mode (ISO 3691-4 or not) and the safety functions it then needs. Decided 2026-10-02: a driverless truck function under ISO 3691-4 (PLP-DEC-001).
- Which donor jack models and steering yokes to support first. Decided 2026-10-02: a survey of three widely sold 27 x 48 in, 2,500 kg jacks with published dimensioned drawings, a Crown PTH 50 series class model, a distributor's standard model such as Uline's and the Harbor Freight Pittsburgh 5,500 lb jack (PLP-DEC-001).
- Are small sites willing to wear a UWB tag and to train every operator as a powered truck operator?
- Typical pallet mass and move distance per shift at real sites, to replace the assumed 60 moves of 40 m.

■ User research and co-design

This design is for workplaces the author does not work in, so requirements come from the people who will use it.

- [] Identify partners: one small warehouse and one maker space, as decided on 2026-09-25 (PLP-DDR-001, D9).
Decided 2026-10-02: the first candidates to approach are Dallas Makerspace and a small food bank or charity warehouse in the Dallas and Fort Worth area, which trials only after the jack maker's written approval under D8; nothing is agreed (PLP-DEC-001)
- [] Observe a full shift: pallet masses, distances, floor condition, ramps, traffic and how often the operator walks back to the jack
- [] Talk to a safety professional and a jack maker about training and modification approval
- [] Revise requirements (REQ) from findings before freezing the design